

Targeted plasma proteomics reveals organ damage signatures of AIDS-related and non-communicable chronic diseases-related deaths in HIV population

Corresponding Author: Professor Na He

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I reviewed with interest the manuscript by Lin et al titled in which the authors sought to identify protein biomarkers associated with HIV death overall and AIDS-related versus NCDs-related death and subsequently develop a protein panel for risk assessment of HIV death. Using a targeted proteomics panel in a nested case-control design, the authors reported associated with both death categories, provided evidence of replication in another population, and showed the discrimination results of the resulting proteomic panel.

The study aims to fill an important gap regarding what is known, at least mechanistically, about HIV-related deaths and provides some useful insights. However, several concerns plague the significance of the message in this report.

Major Comments

1. Study Design and Methods

A. Line 362- 373: Selection bias may be a potential concern here, as less than 6% of the original CHART cohort is included in this report. How do the baseline characteristics of the selected 440 patients compare to the characteristics of the overall population? Could patients have been matched with treatment in addition to age and sex?

B. Line 374: I may have missed it, but the Methods section of the report does not clearly explain how the 127 patients from the validation cohort were selected from the study population. Providing additional details on this selection process would strengthen the transparency of the study methodology.

C. Further information on how the causes of death were ascertained in the study would help to clarify the findings related to mortality.

D. The authors rightly point to the role of Antiretroviral therapy on mortality, however adjustments for the duration or type of ART use are conspicuously missing from the models and could potentially further enhance the models' accuracy. This is particularly important since several ART either act directly on viral or host proteins or indirectly through transcription. It would be important to have additional information on the ART types received and their impact on protein expression in these patients.

E. Adjusting protein expression/association for renal function in a study like this is important as several studies have reported the impact of renal function on protein expression due to the kidney's role in filtering and inflammatory response.

2. Results and Discussion

A. Line 421: The value of the reported differentially expressed organ damage proteins between HIV deaths and survivors (volcano) is not clear in the report. Were these results relevant to the LASSO selection?

B. The terms "Traditional model" and "Traditional biomarkers" are used throughout the manuscript, but a brief introduction or reference to these terms would improve clarity for the reader.

C. Line 254-255: The statement " The expression of four proteins RARRES1, EDIL3, CA12, and CA14 were significantly" is not clear from the Results. Also, the statement implies causality which is not a possible inference from this report. A rephrasing might help to reflect the observational nature of these findings.

D. Discussion section: The emphasis on cancer-related deaths throughout the section seems to derail from the focus of the current report or the aims introduced. (For example, Line 330-331). Consider aligning the Discussion more closely with the

study's objectives and main findings.

3. Interpretation of Results:

The current Discussion section fails to adequately discuss the results but instead focuses on a literature review of the protein findings mostly unrelated to the focus of this current report. Also, there are several unsubstantiated statements that imply causality (Line 280-281, 298 for example) scattered throughout the results section.

Minor

1. Check for consistent use of abbreviations versus words (SIRT5/Sirtuin)
2. Check spelling error in Table 1 "Collage"

Reviewer #2

(Remarks to the Author)

Targeted plasma proteomics reveals organ damage signatures of AIDS- related versus non-communicable chronic diseases-related deaths in HIV population

Review

This manuscript by Lin and colleagues examined 92 soluble proteins selected based on the link to organ damage on 440 cases/controls (people with or without HIV) to determine predictors of end stage organ dysfunction and mortality in persons living with HIV who were enrolled from the Comparative HIV and Aging Research in Taizhou (CHART) Cohort in China that served as a training cohort. A validation cohort of 127 person with HIV.

The key findings are the authors studies 82 proteins (with valid measures) have identified unique proteins and a prediction model that distinguished (13 proteins) HIV deaths from HIV survivors as well as (7) NCDs-related and (2) AIDS-related deaths.

The identification of novel proteins for AIDS-related deaths with, CA14 validated in a separate cohort and new proteins for NCD-related deaths with RARRES1, CA12, and PTN validated in the validation cohort. These markers correlated with traditional biomarkers of NCD in PWH. Finally in pathway analysis in PWH one-carbon metabolic process and response to hypoxia were linked to death. There is a need to identify predictors of HIV associated disease mortality and those at high risk and provide appropriate clinical interventions. Diagnostic tools are not in clinical practice and with the advent of omics and new biomarker technologies and the ability assess an unbiased large proteomic profile with limited sample volumes is emerging in advancing this field.

The report is appropriately written, to infer the results presented and commend the authors for the effort accessing a validation group for the predictive analyses, addressing confounders and appropriate statistical methods with appropriate prediction corrections. The authors also adequately cover the key references in the field noting the limited data. The manuscript provides important results to the field on improved predictive biomarkers of HIV mortality and those markers identified such as Carbonic anhydrases - CA12, CA14 and were consistent and potentially biologically relevant as outlined in the discussion. The authors suggest blood oxygen saturation and pH value are critical markers of end-stage disease in PWH that could be implemented clinically based on the data presented.

The authors should address the following shortcomings and suggestions for improvement as outlined below:

Major Comments:

1. In the abstract on Line 54 is unclear which markers were predictive
2. The limited pre-selection of organ damage linked markers has limited the opportunity to identify novel predictors given the breadth of the O-Link proteome platform and a missed opportunity. This has limited the enthusiasm for the findings.
2. Line 54 "hysteritic" spelling error
3. Number of deaths in the validation cohort n=5 is rather small to support the findings though this is acknowledged and would be cautious in the overinterpretation of the findings related to HIV death.
- Line 74 & 97 > CD4 Cells. Should be corrected to CD4 T cells
3. The authors should use person language eg: healthy controls to People without HIV in line with NIAID language guidelines. (eg: "HIV patients" ; "HIV infected")
4. Any unique findings based on gender?
5. The discussion is disjointed with limited cohesion of the findings.

Minor comments:

1. Can the Authors comment on Substance use or if this information was on a questionnaire – Line 386 Data Collection – and if this was included as a confounder in the analyses.

Reviewer #3

(Remarks to the Author)

- The largest weakness of this manuscript is their analytical approach. Throughout the study (e.g., lines 42, 89 and 241), the authors implied that they seeking to find unique proteomics signatures that dichotomize death by AIDS versus non-AIDS cases (i.e. noncommunicable chronic diseases, NCDs) among people living with HIV. However, the figures and tables were generating comparing AIDS-related death or NCDs-related death with the HIV survivors' group. Depending on these differential expression analyses, the conclusions could have been totally different from the ones withdrawn by the authors. The paper would have benefitted by including an initial schema (e.g., a Venn's diagram) pointing to common proteins found when all the groups are compared against each other (AIDS deaths, NCDs deaths, HIV survivors and HIV negatives).
- The paper fails to describe potential confounders. The reviewers wonder whether the time of plasma isolation as well as the preservative (e.g., EDTA, heparin, etc.) could have introduced methodological artifacts when analyzing proteomics signatures in people with HIV who died versus those that survived. The longer the post-mortem period has been shown to affect the proteome composition (PMID: 37020120). Additionally, the definition of "early" is unclear. The authors do not present data or discuss the time between plasma collection and death. At most, this is 5 years for this study (based on the study enrollment described in line 369). In the discussion, the authors state that protein expression was significantly altered "right before" AIDS or NCD related death (line 255). Together with no description of relevant interventions, it is difficult for the reader to understand the significance of the biomarkers identified in this study.
- The statistical approaches chosen were not very clear. For example, in Table 1, the authors showed statistical differences in demographic, epidemiological and clinical characteristics and reported three P-values. The right test should have been ANOVA followed by a post-hoc test. On the same note, the tables include data that was not statistically significant after FDR correction and adjustment for confounding variables. This made the data unnecessarily difficult for the reader to interpret. Furthermore, in the Statistical Analysis section (starting at line 416), the authors do not specify which sociodemographic characteristics were normally distributed or skewed.
- Some of the statements made in the introduction of the paper are directly contradicted in the results and discussion section. For example, Introduction sentence starting line 75: "traditional biomarkers of NCDs...are challenging in predicting mortality of HIV population" and Discussion sentence starting line 251: "the selected proteins displayed significant associations with traditional biomarkers of NCDs, indicating their potential...for predicting mortality among PWH."
- The manuscript discussion is shallow and confusing. It provides definitions of the protein biomarkers identified, but little insight into potential interesting mechanisms implications for the field. The authors state that "biomarkers differentiating AIDS and NCD related deaths could help early and precise prediction and intervention of death for PWH" (lines 42 and 43). However, they never make this comparison, and they do not describe the interventions that could be taken. Why would a biomarker differentiating AIDS and NCD deaths be useful when these two causes of death are mutually exclusive?
- It is not clear why the validation cohort encompassing 127 samples had significantly lower sensitivity than the training datasets. Supplemental Table 1 is missing.
- Some of the proteins described in line 201 participate in neuroblast proliferation pathway that is functional in the central nervous system. What is the biological relevance in blood? The authors did not address this issue related to biological compartmentalization.
- The association between the level of expression proteins from the same one-carbon metabolism (CA12 and CA14) with NCD-related or AIDS-associated deaths have opposite directions (Table 3). Likewise, the direction of correlation between these two markers and clinical parameters (HIV duration, ART duration) is also in the opposite direction. The authors did not explain this discrepancy and discuss any potential mechanism for this observation.
- The discussion of the paper did not allude to other proteomics study done in people living with HIV; it merely enumerates the function of the differentially expressed proteins without providing a big picture. For example, the authors did not mention the recently published paper on Nature Communications by Victor Riitho et al (PMID: 38944653). This paper also addressed protein biomarkers and mortality in PLWH. Furthermore, it was not discussed previous studies addressing plasma proteomics signatures and all-cause mortality in HIV negative individuals, such as the study by Ulfarsson et al (PMID: 34145379).

Minor points:

- The paper fails to mention that the proteomics signatures are from plasma.
- Figure 2 is not explained, and it should be included in the supplement, after the appropriate interpretations.
- Figure 5 is trimmed.

Reviewer #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Reviewer #2

(Remarks to the Author)

The authors have adequately responded to all comments with the exception of one additional edit required in reference to first person language on Line 111: (shown in Table 2. For HIV patients) should be updated.

Reviewer #3

(Remarks to the Author)

I thank the authors for their thorough and careful reply to all of my concerns and comments.

The authors fully addressed all my points in their revised manuscript.

The only minor error that still is present in the manuscript is in the statistical description.

Because the distribution of triglycerides is skewed but because there are three comparison groups, the right test should be Kruskal Wallis (3 or more groups) and not Mann Whitney U test (2 groups).

This should be changed in Lines 533-537 and also at the footnote of Table 1. Maybe the authors want to check if the p-value was calculated accordingly.

Reviewer #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

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Authors' Point-by-point Responses to The Reviewers' Comments

Reviewer #1 (Remarks to the Author):

I reviewed with interest the manuscript by Lin et al titled in which the authors sought to identify protein biomarkers associated with HIV death overall and AIDS-related versus NCDs-related death and subsequently develop a protein panel for risk assessment of HIV death. Using a targeted proteomics panel in a nested case-control design, the authors reported associated with both death categories, provided evidence of replication in another population, and showed the discrimination results of the resulting proteomic panel.

The study aims to fill an important gap regarding what is known, at least mechanistically, about HIV-related deaths and provides some useful insights. However, several concerns plague the significance of the message in this report.

Major Comments

1. Study Design and Methods

A. Line 362- 373: Selection bias may be a potential concern here, as less than 6% of the original CHART cohort is included in this report. How do the baseline characteristics of the selected 440 patients compare to the characteristics of the overall population? Could patients have been matched with treatment in addition to age and sex?

Response: We thank the reviewer for this important comment about potential selection bias in our study. We would like to clarify that our research follows a nested case-control design where we first identified death cases and then selected controls from the cohort through matching. Our study design does not aim to obtain estimates from the entire cohort but, rather, to set a comparative analysis between cases and matched controls.

Given this study design, we first identified death cases from the cohort and then used propensity score matching to select controls based on age and sex. Propensity score matching provided an appropriate comparison between cases and controls and ensures the internal validity of the results. We have added the following sentences to the methods section to better clarify the selection process:

“Propensity score matching was used to select individuals for control groups based on age and sex, ensuring comparability between HIV deaths and controls. Two control groups including 162 HIV-positive survivors and 152 HIV-negative individuals were eventually selected through matching.” (Line 485-489/Page 20-21)

Also, we understand that the treatment plan may be a potential source of bias in our analysis. We recognize that the treatment plan between the death cases group and the HIV positive control

group differed for several medications, including Tenofovir disoproxil, Zidovudine, Dolutegravir, Nevirapine, and Efavirenz. Tenofovir disoproxil is known to be associated with nephrotoxicity, which could influence the proteomic results¹. However, in the current study, we also have a healthy control group that did not receive any treatment. Since untreated controls were included, we only matched for age and sex to avoid potential confusion. As treatment plans can be a source of bias, we have now controlled for both treatment plan and treatment duration in our analysis model. The new results are provided in *Table 3. Logistic regression analysis of selected proteins among different comparison groups* and *Supplemental table 1. Logistic regression analysis of selected proteins among different comparison groups in validation cohort*

B. Line 374: I may have missed it, but the Methods section of the report does not clearly explain how the 127 patients from the validation cohort were selected from the study population. Providing additional details on this selection process would strengthen the transparency of the study methodology.

Response: We appreciate the reviewer's comment about the selection process. In the validation cohort, we first identified 52 death cases and then selected 86 HIV-positive controls using propensity score matching according to age and sex as we had done in the discovery cohort. During the analysis, 11 death cases were excluded because their causes of death were unrelated to both HIV and NCDs, leaving 41 death cases for the final analysis. We have added the following in the methods section to better describe the selection process of the validation of participants:

“Consistent with the CHART cohort, controls for the validation phase were also selected using propensity score matching based on age and sex. A total of 86 HIV survivors, matched for age and sex, were included in the analysis.” (Line 497-500/Page 21)

C. Further information on how the causes of death were ascertained in the study would help to clarify the findings related to mortality.

Response: The mortality data used in our study were obtained from the Taizhou Center for Disease Control and Prevention (exploration samples) and the Dehong Prefecture Center for Disease Control and Prevention (validation samples). The causes of death were ascertained by these centers based on detailed medical records, which include clinical diagnoses and death record. We added detailed information in the supplemental materials:

“The causes of death in our study were ascertained using detailed mortality data obtained from the Taizhou Center for Disease Control and Prevention (exploration samples) and the Dehong Prefecture Center for Disease Control and Prevention (validation samples). The Dehong CDC determined the causes of death based on medical records, which included clinical diagnoses and official death records. These records were carefully reviewed and validated to ensure accuracy. Deaths were then categorized into AIDS-related and NCD-related deaths based on the primary cause listed in the records.” (Supplemental materials)

D. The authors rightly point to the role of Antiretroviral therapy on mortality, however adjustments for the duration or type of ART use are conspicuously missing from the models and could potentially further enhance the models' accuracy. This is particularly important since several ART either act directly on viral or host proteins or indirectly through transcription. It would be important to have additional information on the ART types received and their impact on protein expression in these patients.

Response: We thank the reviewer for this suggestion. We agree with that the type and duration of ART should be considered in our analysis as we described in the response for comment A. We now included adjustments for ART medicine type in our logistic regression model for both primary analysis and validation analysis. This adjustment controls for potential bias from the influence of ART medications on protein expression in PWH. Besides, in our study, the participants started to receive ART when they were diagnosed. Therefore, we have already adjusted for the duration of ART through the adjustment for HIV duration in our analysis.

We have added the following statement to the Supplemental materials to provide detailed information about how ART medications were included in the model:

“Data of ART usage was extracted from the CDC’s records. Each ART medication was treated as an individual binary variable, coded as “1” if a patient had used the medication and “0” otherwise. This includes medicines as Lamivudine, Dolutegravir, Zidovudine, lopinavir, Nevirapine, Tenofovir disoproxil, Tenofovir Alafenamide, Bictegravir, Emtricitabine and Elvitegravir.” (Data collection and measurements). The updated results are shown in **Table 3. Logistic regression analysis of selected proteins among different comparison groups and Supplemental table 1. Logistic regression analysis of selected proteins among different comparison groups in validation cohort.**

E. Adjusting protein expression/association for renal function in a study like this is important as several studies have reported the impact of renal function on protein expression due to the kidney's role in filtering and inflammatory response.

Response: This is a great suggestion. We agree that renal function is important and should be adjusted for in the analysis due to its filtering and inflammatory response mechanisms². We have now included eGFR into our logistic regression models for both primary analysis and validation analysis, and the results of this adjustment are now included in the new table provided in the revised manuscript.

Please see: *Upon adjustment for age, sex, BMI, tobacco use, regular alcohol use, eGFR, ART type and HIV duration... (Line 135-136, 153-154, 251-252/Page8, 9, 13)*

The updated results are shown in **Table 3. Logistic regression analysis of selected proteins among different comparison groups and Supplemental table 1. Logistic regression analysis of selected proteins among different comparison groups in validation cohort**

2. Results and Discussion

A. Line 421: The value of the reported differentially expressed organ damage proteins between HIV deaths and survivors (volcano) is not clear in the report. Were these results relevant to the LASSO selection?

Response: The results presented in the volcano plot reflect the proteins that were differentially expressed between HIV deaths and survivors based on the t-test analysis and 54 proteins were differentially expressed between HIV deaths and survivors. For the LASSO regression, we included all proteins in the analysis, not just those identified as significant in the volcano plot. Consequently, some proteins identified by LASSO as predictive were not statistically significant in the univariate t-test analysis used for the volcano plot. We compared the proteins selected by the two methods and found that a subset of proteins identified by LASSO regression were not significant in the volcano plot. These non-significant proteins were excluded from further logistic regression analysis. The volcano plot provides an overview of differential expression, LASSO focuses on identifying the most predictive biomarkers for the risk assessment of HIV deaths.

:

B. The terms "Traditional model" and "Traditional biomarkers" are used throughout the manuscript, but a brief introduction or reference to these terms would improve clarity for the reader.

Response: We thank the reviewer for pointing this out. The term "traditional model" refers to the integration of NCDs biomarkers such as age, sex and cIMT with HIV biomarkers such as CD4 T cells counts as we compared in the ROC curves. To improve clarity for the reader, we have now revised *"NCDs are more likely to occur and progress more rapidly in PWH than in the general population, and biomarkers of NCDs such as carotid intima-media thickness (cIMT) used in general population may not reliably predict disease progression in PWH, as cIMT differs significantly between general population and PWH. Traditional models that integrate NCDs biomarkers with CD4 T cell counts are insufficient in predicting mortality of PWH which require supplementation with novel biomarkers."* (Line 78-83/Page5) in the introduction.

C. Line 254-255: The statement "The expression of four proteins RARRES1, EDIL3, CA12, and CA14 were significantly" is not clear from the Results. Also, the statement implies causality which is not a possible inference from this report. A rephrasing might help to reflect the observational nature of these findings.

Response: We agree that the original phrasing may imply a causal relationship, which would be an inappropriate interpretation of the results. We have revised the statement based on the suggestion and provide the information in the results section:

"The expression of four proteins RARRES1, EDIL3, CA12, and CA14 were significantly associated with NCD-related death and AIDS-related death." (Line 376-377/Page 16)

and the results are shown in:

"The six proteins were ranked from the smallest FDR-corrected p-value as RCOR1, CA14, CA12, RARRES1, EDIL3, SERPINA9 (Table 3)." (Line 156-158/Page 7)

D. Discussion section: The emphasis on cancer-related deaths throughout the section seems to derail from the focus of the current report or the aims introduced. (For example, Line 330-331). Consider aligning the Discussion more closely with the study's objectives and main findings.

Response: We agree with the reviewer's comment regarding the overemphasis on cancer-related deaths in the discussion section. We have revised the discussion section to focus more on the potential mechanisms related to NCDs and AIDS-related deaths. These mechanisms include abnormal differentiation of myeloid lineage cells, chronic inflammation, oxidative stress, and cellular damage. As discussed in the revised manuscript:

“Previous studies demonstrated the involvement of RCOR1 in hematopoiesis through regulating the differentiation of megakaryocyte-erythroid progenitor cells. Additionally, prior studies also shown that HIV infection could induce abnormal myeloid lineage cells differentiation and certain hematopoietic progenitor cells can serve as latent cellular reservoirs.....In pressure overload-induced hypertrophic cardiomyopathy (HCM), PTN secreted by cardiac fibroblasts activates the PTN-SDC4 pathway, promoting myocardial fibrosis and macrophage-mediated inflammation. Furthermore, elevated PTN levels have been identified in patients with inflammatory diseases. ”
(Line 276-370/Page 14-18)

3. Interpretation of Results:

The current Discussion section fails to adequately discuss the results but instead focuses on a literature review of the protein findings mostly unrelated to the focus of this current report. Also, there are several unsubstantiated statements that imply causality (Line 280-281, 298 for example) scattered throughout the results section.

Response: We fully agree with the reviewer's comment regarding the discussion section. As described in our previous response, we have revised the discussion to focus more on the potential mechanisms related to NCDs and AIDS-related deaths, particularly emphasizing on inflammation.

In addition, we have added a section in the revised manuscript comparing our results with findings from previous studies. Several significant proteins identified in our study are related to inflammation, aligning with findings from the REALITY trial, which identified multiple inflammatory biomarkers as predictors of mortality in people with HIV (PWH) in sub-Saharan Africa. We also compared our results with a previous proteomics study on mortality in the general population. As we added in the manuscript *“Several significant proteins identified in this study are related to inflammation. Aligning with the previous findings from the REALITY trial, which identified multiple inflammatory biomarkers as mortality predictors for PWH in sub-Saharan Africa. PTN, identified in our study, is linked to IFN- γ as discovered in this previous study, underscoring the shared role of inflammation in advanced HIV. These findings highlight the critical role of inflammation-related pathways in PWH. Compared to a previous study investigating proteomics in the general population, only PTN was identified in both studies. The distinction underscores differences in protein profiles between healthy individuals and PWH.*

These findings suggest the potential distinct pathophysiological mechanisms driving mortality in these two populations and emphasize the importance of conducting HIV-specific studies to better understand and address the unique drivers of death among PWH. (Line 410-422/Page 20-21)". We have also refined the text to avoid any unsubstantiated statements implying causality.

Minor

1. Check for consistent use of abbreviations versus words (SIRT5/Sirtuin)

Response: Revised as suggested

2. Check spelling error in Table 1 “ Collage”

Response: Revised as suggested.

Reviewer #2 (Remarks to the Author):

Targeted plasma proteomics reveals organ damage signatures of AIDS- related versus non-communicable chronic diseases-related deaths in HIV population

Review

This manuscript by Lin and colleagues examined 92 soluble proteins selected based on the link to organ damage on 440 cases/controls (people with or without HIV) to determine predictors of end stage organ dysfunction and mortality in persons living with HIV who were enrolled from the Comparative HIV and Aging Research in Taizhou (CHART) Cohort in China that served as a training cohort. A validation cohort of 127 person with HIV.

The key findings are the authors studies 82 proteins (with valid measures) have identified unique proteins and a prediction model that distinguished (13 proteins) HIV deaths from HIV survivors as well as (7) NCDs-related and (2) AIDS-related deaths.

The identification of novel proteins for AIDS-related deaths with, CA14 validated in a separate cohort and new proteins for NCD-related deaths with RARRES1, CA12, and PTN validated in the validation cohort. These markers correlated with traditional biomarkers of NCD in PWH. Finally in pathway analysis in PWH one-carbon metabolic process and response to hypoxia were linked to death. There is a need to identify predictors of HIV associated disease mortality and those at high risk and provide appropriate clinical interventions. Diagnostic tools are not in clinical practice and with the advent of omics and new biomarker technologies and the ability assess an unbiased large proteomic profile with limited sample volumes is emerging in advancing this field.

The report is appropriately written, to infer the results presented and commend the authors for the effort accessing a validation group for the predictive analyses, addressing confounders and appropriate statistical methods with appropriate prediction corrections. The authors also adequately cover the key references in the field noting the limited data. The manuscript provides

important results to the field on improved predictive biomarkers of HIV mortality and those markers identified such as Carbonic anhydrases - CA12, CA14 and were consistent and potentially biologically relevant as outlined in the discussion. The authors suggest blood oxygen saturation and pH value are critical markers of end-stage disease in PWH that could be implemented clinically based on the data presented.

The authors should address the following shortcomings and suggestions for improvement as outlined below:

Major Comments:

In the abstract on Line 54 is unclear which markers were predictive

Response: Thank you to the reviewer for pointing this out. The sentence has been revised as: *“The adjusted protein panels consisted of significantly associated proteins selected through both LASSO regression model and logistic regression model well predicted NCDs-related death (AUC=0.970) and AIDS-related death (AUC=0.960) in PWH, respectively.”* (Line 53-57/Page 3)

The limited pre-selection of organ damage linked markers has limited the opportunity to identify novel predictors given the breadth of the O-Link proteome platform and a missed opportunity. This has limited the enthusiasm for the findings.

Response: We acknowledge the limitation of using targeted pre-selected organ damage-linked proteomics. This study’s primary aim is to investigate organ damage-related proteins associated with deaths in PWH. Hence, we decided to only focus on a subset of markers instead of conducting untargeted proteomics. We recognize that this approach may have restricted the discovery of novel predictors, and we discussed this as a limitation in the discussion section. Please see: *First, only 92 proteins were measured. However, the proteins measured in our study were specifically related to organ damage and may have a more direct relationship with mortality.* (Line 452-454/Page 22)

Line 54 “hysteritic” spelling error

Response: Revised as suggested.

Number of deaths in the validation cohort n=5 is rather small to support the findings though this is acknowledged and would be cautious in the overinterpretation of the findings related to HIV death.

Response: We acknowledge the limitation of the small number of deaths in the validation cohort, especially the AIDS-related deaths. This limited sample size was the primary reason the validation cohort was used only for validation purposes rather than discovery. The small sample size may also have contributed to certain proteins not being validated in the validation cohort. However, the consistency of three proteins validated in the validation phase still strengthens our findings. We recognized this important limitation and we have discussed this as a limitation in the discussion section in the revised manuscript.

Please see: *Second, we conducted a subsequent validation analysis; however, we were unable to validate all the significant findings from the primary analysis in the validation sample due to the insufficient call rate of several proteins and insufficient death cases in the validation sample. The small number of AIDS-related deaths led to overparameterized models which resulting in uninterpretable outcomes.* (Line 454-458/Page 22)

Line 74 & 97 > CD4 Cells. Should be corrected to CD4 T cells

Response: Revised as suggested.

The authors should use person language eg: healthy controls to People without HIV in line with NIAID language guidelines. (eg:“HIV patients” ; “HIV infected”)

Response: We have revised the manuscript to use affirming and non-stigmatizing language.

Any unique findings based on gender?

Response: The majority of participants in our sample is male as shown in Table 1 which limits the power to draw conclusions about gender-specific differences. Therefore, gender was not a significant covariate in our model, and we did not identify any unique patterns based on gender.

The discussion is a disjointed with limited cohesion of the findings.

Response: We agree that the discussion can be improved for better cohesion. We have revised the discussion to discuss more on the potential mechanisms related to NCDs and AIDS-related deaths which align with our findings.

In addition, we also added a comparison of our results with previous studies in the discussion. Several significant proteins identified in our study are related to inflammation, aligning with findings from the REALITY trial, which identified multiple inflammatory biomarkers as predictors of mortality in people with HIV (PWH) in sub-Saharan Africa. We also compared our results with a previous proteomics study on mortality in the general population.

Minor comments:

Can the Authors comment on Substance use or if this information was on a questionnaire – Line 386 Data Collection – and if this was included as a confounder in the analyses.

Response: Thank you for your comment. We do have questions about substance use in our questionnaire. However, very few participants reported substance use in our sample. Only three participants answered "yes" to these questions. Due to the low prevalence, we did not include substance use as a confounder in our analysis.

Reviewer #3 (Remarks to the Author):

- The largest weakness of this manuscript is their analytical approach. Throughout the study (e.g., lines 42, 89 and 241), the authors implied that they seeking to find unique proteomics signatures that dichotomize death by AIDS versus non-AIDS cases (i.e. noncommunicable chronic diseases, NCDs) among people living with HIV. However, the figures and tables were generating comparing AIDS-related death or NCDs-related death with the HIV survivors' group. Depending on these differential expression analyses, the conclusions could have been totally different from the ones withdrawn by the authors. The paper would have benefitted by including an initial schema (e.g., a Venn's diagram) pointing to common proteins found when all the groups are compared against each other (AIDS deaths, NCDs deaths, HIV survivors and HIV negatives).

Response: Thank you for your careful review of our manuscript. To clarify, our main goal was not to dichotomize deaths into AIDS-related versus non-AIDS cases but to investigate proteomic signatures associated with mortality among people with HIV and identifying unique proteins linked specifically to AIDS-related and NCD-related deaths. We realize that some statements in the manuscript (e.g., lines 42, 89, and 241) may provide implications to this, but they were unintentional. To address this, we have revised the manuscript to more adequately reflect our objectives. Additionally, we included a Venn diagram showing both shared and unique proteins among AIDS-related deaths, NCD-related deaths, and cancer-related deaths for a better presentation of the findings.

Please see: *Protein biomarkers differentiating both AIDS-related and NCDs-related deaths from PWH may help early and precise risk prediction and intervention of death for PWH.* (Line 42-44/Page 3)

And

To fill this gap, we measured plasma levels of 92 organ damage related proteins using the Olink PEA technique and examined their associations with overall HIV-related mortality, as well as their specific links to AIDS-related and NCD-related deaths. (Line 92-94/Page 6)

And

This is the first study of profiling proteomic signatures of NCDs-related death and AIDS-related death in HIV population. (Line 259-260/Page 13)

And

Taken together, our proteomics analysis identified a novel set of proteins associated with HIV-related deaths compared to HIV survivors, including proteins specifically linked to NCD-related or AIDS-related deaths, with four of these proteins successfully validated in an independent sample. (Line 462-465/Page 23)

And

To visualize the overlap of organ damage proteins associated with AIDS-related, NCDs-related, and cancer-related deaths, a Venn diagram was created (Fig 3). As the Venn diagram shown, RCOR1 and SERPINA9 were specific to AIDS-related deaths, while GALNT10, VEGFC, and PTN were specific to NCDs-related deaths. Shared proteins between NCDs-related and cancer-related

deaths included SIRT5, PPM1B, and PSMA1. Notably, RARRES1, EDIL3, CA12, and CA14 were shared among AIDS-related, NCDs-related, and cancer-related deaths. (Line 160-167/Page 9)
And

Fig 3. Venn diagram of proteins associated with AIDS-related, NCDs-related, and cancer-related deaths.

- The paper fails to describe potential confounders. The reviewers wonder whether the time of plasma isolation as well as the preservative (e.g., EDTA, heparin, etc.) could have introduced methodological artifacts when analyzing proteomics signatures in people with HIV who died versus those that survived. The longer the post-mortem period has been shown to affect the proteome composition (PMID: 37020120). Additionally, the definition of “early” is unclear. The authors do not present data or discuss the time between plasma collection and death. At most, this is 5 years for this study (based on the study enrollment described in line 369). In the discussion, the authors state that protein expression was significantly altered “right before” AIDS or NCD related death (line 255). Together with no description of relevant interventions, it is difficult for the reader to understand the significance of the biomarkers identified in this study.

Response: We thank the reviewer for pointing this out. All plasma EDTA samples were collected at baseline when participants were enrolled in the CHART cohort and were processed using a uniform protocol for plasma isolation. The samples were stored at -80°C without any prior thawing to minimize degradation or preservation-related issues. Additionally, we evaluated the time between sample collection and Olink analysis; the results showed no significant difference between the HIV death group (median: 64.43 months; IQR: 53.78-66.97) and the HIV-positive control group (median: 64.7 months; IQR: 63.91-65.81; P = 0.119). For the HIV death group, the time between sample collection and death was evaluated and the median time was 21.67 months (IQR: 1.6 – 37.5 months). We have deleted the term “right before” in the manuscript for accuracy. We added “*Among HIV deaths, 27.78% (35/126) died of AIDS-related diseases whereas 72.22% (91/126) died of non-AIDS-related NCDs. The median time between plasma collection and death was 21.67 months (IQR: 1.6 – 37.5 months).*” (Line 108-110/Page 7) in the results section to provide information about the time between plasma collection and death.

- The statistical approaches chosen were not very clear. For example, in Table 1, the authors showed statistical differences in demographic, epidemiological and clinical characteristics and reported three P-values. The right test should have been ANOVA followed by a post-hoc test. On the same note, the tables include data that was not statistically significant after FDR correction and adjustment for confounding variables. This made the data unnecessarily difficult for the reader to interpret. Furthermore, in the Statistical Analysis section (starting at line 416), the authors do not specify which sociodemographic characteristics were normally distributed or skewed.

Response: We appreciate this valuable feedback. We have revised Table 1 to provide only the meaningful comparison of each sociodemographic characteristic of the three groups: one p-value

from Chi-squared tests (for comparison of frequencies/percentages), ANOVA (for comparison of means), or Mann Whitney U test (for comparison of medians). We also removed data that were not statistically significant after FDR correction and adjustment for confounding variables from Table 3. We have confirmed that the continuous variables in Table 1, including total cholesterol, HDL, and LDL are normally distributed. Triglycerides were skewed and thus Mann Whitney U test was used for comparisons between the three groups. We have modified the manuscript to include this information:

“Comparisons of sociodemographic characteristics, stratified by HIV serostatus and death status, were performed using chi-square test for categorical variables, analysis of variance (ANOVA) for normally distributed continuous variables (total cholesterol, HDL, and LDL), and Mann-Whitney U test for continuous variables with skewed distributions (triglycerides).” (Line 533-537/Page 26)

We have also added a footnote under Table 1:

“Data are presented as counts (percentage of total), mean +/- SD, or median (IQR). P-values were obtained from chi-square test for categorical variables, analysis of variance (ANOVA) for normally distributed continuous variables (total cholesterol, HDL, and LDL), and Mann-Whitney U test for continuous variables with skewed distributions (triglycerides).”

- Some of the statements made in the introduction of the paper are directly contradicted in the results and discussion section. For example, Introduction sentence starting line 75: “traditional biomarkers of NCDs...are challenging in predicting mortality of HIV population” and Discussion sentence starting line 251: “the selected proteins displayed significant associations with traditional biomarkers of NCDs, indicating their potential...for predicting mortality among PWH.”

Response: We have revised the manuscript to ensure consistency. We have clarified in the introduction that traditional biomarkers have limitations in predicting mortality in PWH and that their integration with novel proteomic markers can enhance prediction. In the discussion, we clarified that our findings may provide support as a complementary role along with traditional biomarkers.

We revised these sentence as: *“NCDs are more likely to occur and progress more rapidly in PWH than in the general population, and biomarkers of NCDs such as carotid intima-media thickness (cIMT) used in general population may not reliably predict disease progression in PWH, as cIMT differs significantly between general population and PWH. Traditional models that integrate NCDs biomarkers with CD4 T cell counts are insufficient in predicting mortality of PWH which require supplementation with novel biomarkers.”* (Line 78-83/Page5)

And

“Furthermore, the selected proteins displayed significant associations with traditional biomarkers of NCDs, demonstrate their potential to complement traditional biomarkers of NCDs and enhance their ability in predicting mortality among PWH.” (Line 270-273/Page 14)

- The manuscript discussion is shallow and confusing. It provides definitions of the protein

biomarkers identified, but little insight into potential interesting mechanisms implications for the field. The authors state that “biomarkers differentiating AIDS and NCD related deaths could help early and precise prediction and intervention of death for PWH” (lines 42 and 43). However, they never make this comparison, and they do not describe the interventions that could be taken. Why would a biomarker differentiating AIDS and NCD deaths be useful when these two causes of death are mutually exclusive?

Response: We appreciate this feedback. We have thoroughly revised the discussion to provide deeper insight into potential mechanisms of interest. These mechanisms include abnormal differentiation of myeloid lineage cells, chronic inflammation, oxidative stress, and cellular damage. As discussed in the revised manuscript *“Previous studies demonstrated the involvement of RCOR1 in hematopoiesis through regulating the differentiation of megakaryocyte-erythroid progenitor cells. Additionally, prior studies also shown that HIV infection could induce abnormal myeloid lineage cells differentiation and certain hematopoietic progenitor cells can serve as latent cellular reservoirs.....In pressure overload-induced hypertrophic cardiomyopathy (HCM), PTN secreted by cardiac fibroblasts activates the PTN-SDC4 pathway, promoting myocardial fibrosis and macrophage-mediated inflammation. Furthermore, elevated PTN levels have been identified in patients with inflammatory diseases.”* (Line 276-370/Page 14-18)

We have also adjusted the discussion to clarify that our objective was not to differentiate between AIDS-related and NCD-related deaths, but to distinguish both AIDS- related and NCD-related deaths from the general population of PWH and also to identify unique proteins associated with each of these causes of death, providing novel insights that could contribute to the understanding of mortality risk and disease progression in PWH. We revised the sentence as: *“The confluence of identified organ damage proteins demonstrated high precision in both NCDs-related death and AIDS-related death among general HIV patients compared to traditional models.”* (Line 439-441/Page 22)

- It is not clear why the validation cohort encompassing 127 samples had significantly lower sensitivity than the training datasets. Supplemental Table 1 is missing.

Response: The primary reason for the lower sensitivity in the validation cohort is the small sample size, particularly for AIDS-related deaths. Out of the 127 validation samples, only 6 cases were attributed to AIDS-related deaths. This limited sample size significantly impacts the sensitivity of our findings in the validation group. Additionally, the validation cohort have different sociodemographic characteristics compared to the training cohort, such as smoking status and years since HIV diagnosis, as shown in Supplemental Table 5. Moreover, some proteins identified in the primary analysis did not achieve a sufficient valid call rate in the validation samples. As a result, we were unable to validate this subset of proteins. We have revised the discussion section to discuss the limitation of the validation samples. We have also fixed the problem with supplemental Table 1.

We added the following to the discussion section to address the limitations of the validation samples:

“Second, we conducted a subsequent validation analysis; however, we were unable to validate all the significant findings from the primary analysis in the validation sample due to the insufficient call rate of several proteins and insufficient death cases in the validation sample. The small number of AIDS-related deaths led to overparameterized models which resulted in uninterpretable outcomes. ” (Line 454-459/Page 22)

- Some of the proteins described in line 201 participate in neuroblast proliferation pathway that is functional in the central nervous system. What is the biological relevance in blood? The authors did not address this issue related to biological compartmentalization.

Response: We thank the reviewer for the valuable feedback. We recognize that while these proteins are associated with the neuroblast proliferation pathway specific in the central nervous system and have no established biological relevance in blood circulation. Their presence in plasma likely reflects other mechanisms unrelated to neuroblast proliferation outside of the CNS. For example, PTN is known to induce myocardial fibrosis and macrophage-mediated inflammation³. We have revised the manuscript to point out that although these proteins were identified as related to the neuroblast proliferation pathway by pathway enrichment analysis, the pathways are only specific to the central nervous system.

We revised this sentence as *“Additionally, VEGFC and PTN were associated with the regulation of neuroblast proliferation which is a pathway specific to the central nervous system.”* (Line 216-218/Page 11-12)

- The association between the level of expression proteins from the same one-carbon metabolism (CA12 and CA14) with NCD-related or AIDS-associated deaths have opposite directions (Table 3). Likewise, the direction of correlation between these two markers and clinical parameters (HIV duration, ART duration) is also in the opposite direction. The authors did not explain this discrepancy and discuss any potential mechanism for this observation.

Response: Specifically, we found that although both CA12 and CA14 are membrane-associated carbonic anhydrase (CA) isoforms and play a role in one-carbon metabolism, their functions appear to be tissue-dependent. The contrasting associations suggest that while both are involved in metabolic regulation, they may respond differently to tissue damage caused by HIV, highlighting the need for further investigation into their distinct roles. We have revised the discussion section to discuss the opposite correlations: *“Interestingly, the association of CA12 and CA14 with both NCD-related and AIDS-associated deaths had an opposite pattern. CA12 and CA14 are normally expressed in different tissues but have very similar primary sequence and number of conserved residues. They are both membrane-associated CA isoforms. The opposite pattern of CA12 and CA14 suggests that although both of them are part of 1C metabolism, their functions might be tissue-dependent and influenced differently with varying tissue damage caused by HIV, which*

warrants further research.” (Line 402-409/Page 20)

- The discussion of the paper did not allude to other proteomics study done in people living with HIV; it merely enumerates the function of the differentially expressed proteins without providing a big picture. For example, the authors did not mention the recently published paper on Nature Communications by Victor Riitho et al (PMID: 38944653). This paper also addressed protein biomarkers and mortality in PLWH. Furthermore, it was not discussed previous studies addressing plasma proteomics signatures and all-cause mortality in HIV negative individuals, such as the study by Ulfarsson et al (PMID: 34145379).

Response: We appreciate the reviewer’s valuable feedback. In the revised manuscript, we have added a section comparing our results with findings from previous studies. Several significant proteins identified in our study are related to inflammation, aligning with findings from the REALITY trial, which identified multiple inflammatory biomarkers as predictors of mortality in people with HIV (PWH) in sub-Saharan Africa. We also compared our results with a previous proteomics study on mortality in the general population. We have added the following in the manuscript:

“Several significant proteins identified in this study are related to inflammation. Aligning with the previous findings from the REALITY trial, which identified multiple inflammatory biomarkers as mortality predictors for PWH in sub-Saharan Africa. PTN, identified in our study, is linked to IFN- γ as discovered in this previous study, underscoring the shared role of inflammation in advanced HIV. These findings highlight the critical role of inflammation-related pathways in PWH. Compared to a previous study investigating proteomics in the general population, only PTN was identified in both studies. The distinction underscores differences in protein profiles between healthy individuals and PWH. These findings suggest the potential distinct pathophysiological mechanisms driving mortality in these two populations and emphasize the importance of conducting HIV-specific studies to better understand and address the unique drivers of death among PWH.” (Line 410-422/Page 20-21).

Minor points:

- The paper fails to mention that the proteomics signatures are from plasma.

Response: We thank the reviewer for pointing this out. To address this concern, we have revised the manuscript to mention that the proteomics signatures are from plasma:

To fill this gap, we measured plasma levels of 92 organ damage related proteins using the Olink PEA technique and examined their associations with both AIDS-related and NCDs-related deaths. (Line 92-94/Page 6)

- Figure 2 is not explained, and it should be included in the supplement, after the appropriate interpretations.

Response: We thank the reviewer for pointing this out. We have relocated Figure 2 as Supplemental

Figure 1 and have explained it in the manuscript: “*A coefficient profile plot and a cross-validated error plot of the LASSO regression model were shown in Supplemental Fig .1A and Supplemental Fig .1B, respectively.*” (129-131/Page 7-8)

• Figure 5 is trimmed.

Response: We thank the reviewer for pointing this out. We have corrected this.

1. Morlat, P., *et al.* Role of traditional risk factors and antiretroviral drugs in the incidence of chronic kidney disease, ANRS CO3 Aquitaine cohort, France, 2004-2012. *PLoS One* **8**, e66223 (2013).
2. Matias-Garcia, P.R., *et al.* Plasma Proteomics of Renal Function: A Transethnic Meta-Analysis and Mendelian Randomization Study. *J Am Soc Nephrol* **32**, 1747-1763 (2021).
3. Pufe, T., Bartscher, M., Petersen, W., Tillmann, B. & Mentlein, R. Pleiotrophin, an embryonic differentiation and growth factor, is expressed in osteoarthritis. *Osteoarthritis Cartilage* **11**, 260-264 (2003).

Authors' Point-by-point Responses to The Reviewers' Comments

Reviewer #2 (Remarks to the Author):

The authors have adequately responded to all comments with the exception of one additional edit required in reference to first person language on Line 111: (shown in Table 2. For HIV patients) should be updated.

Response: Thank you for your feedback. We have revised the sentence on Line 111 to use person-first language.

Reviewer #3 (Remarks to the Author):

I thank the authors for their thorough and careful reply to all of my concerns and comments.

The authors fully addressed all my points in their revised manuscript.

The only minor error that still is present in the manuscript is in the statistical description.

Because the distribution of triglycerides is skewed but because there are three comparison groups, the right test should be Kruskal Wallis (3 or more groups) and not Mann Whitney U test (2 groups). This should be changed in Lines 533-537 and also at the footnote of Table 1. Maybe the authors want to check if the p-value was calculated accordingly.

Response: Thank you for your feedback. We have checked the statistical methods used and confirmed that the test we applied for comparing triglycerides was Kruskal Wallis test. We have revised the text on Lines 533–537 and updated the footnote in Table 1 accordingly.